Engineering Statistics

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Chapter 2: Data Summary and Visualization

Chapter Goals: Numerical Summaries

- Measuring the center of data
  - Mean
  - Median
- Measuring the spread of data
  - Variance and standard deviation
  - Five-number summary
  - Interquartile range (IQR)

Chapter Goals: Data Visualization

- Categorical variables
  - Pie charts
  - Bar charts
- Numerical variables
  - Histograms
  - Boxplots
- Scatterplots and Pearson correlation

Measuring Center: The Mean

The most common measure of center is the ordinary arithmetic average, or mean.

For observations $x_1, x_2, \ldots, x_n$, the sample mean is

$$\bar{x} = \frac{x_1 + x_2 + \cdots + x_n}{n} = \frac{1}{n} \sum_{i=1}^{n} x_i.$$

Example: Spider Silk Yield Stress

Spider silk is one of the strongest known materials on a weight basis. A study examined the mechanical properties of silk from 21 female golden orb weavers (*Nephila clavipes*). Yield stress, measured in megapascals (MPa), is the force per unit area required to reach permanent deformation.

164.0	478.7	251.3	351.7	173.0	448.9	300.6
362.0	272.4	740.2	329.0	327.2	270.5	332.1
288.8	176.1	282.2	236.1	358.2	270.5	290.7

1. Find the mean yield stress.
2. How many of the 21 spiders had yield stress below the mean?

R Code Illustration: Mean

```
setwd("C:/your directory")
```

Ask Gemini: Write R code to read in the CSV file `SpiderSilk.csv`, located in the same directory.

```
# Read the CSV file
spider_data <- read.csv("SpiderSilk.csv")
# View the first few rows
head(spider_data)
```

Ask Gemini: Write R code to find the mean of the variable `YieldStress` in `spider_data`.

```
# Calculate the mean of YieldStress
mean_yield <- mean(spider_data$YieldStress)
# Print the result
print(mean_yield)
```

R Outputs

Ask Gemini: Write R code to find the mean of variable `YieldStress` of the `spider_data`.

```
# Calculate the mean of YieldStress
mean_yield <- mean(spider_data$YieldStress)
# Print the result
print(mean_yield)
[1] 319.2476
```

Ask Gemini: Write R code: How many of the 21 spiders had a yield stress less than the mean?

```
# 1. Calculate the mean
mean_yield <- mean(spider_data$YieldStress, na.rm = TRUE)
# 2. Count how many values are less than the mean
count_less <- sum(spider_data$YieldStress < mean_yield,
              na.rm = TRUE)
# 3. View the result
count_less
[1] 12
```

Measuring Center: The Median

The median is the midpoint of a distribution: half of the observations are smaller and half are larger.

1. Arrange all observations from smallest to largest.
2. If n is odd, the median is the center observation.
3. If n is even, the median is the mean of the two center observations.

Example: Aleppo Pine Needle Lengths

Examining nature closely reveals surprising variability. The needles in a given pine tree species are not all the same size. Thus, the distribution of lengths characterizes the species.

Here are the lengths (cm) of 11 needles taken at random from different parts of several Aleppo pine trees located in Southern California:

$$7.2,\ 7.6,\ 8.5,\ 8.7,\ 9.0,\ 9.3,\ 10.2,\ 10.9,\ 11.3,\ 12.1,\ 12.8$$

Question: Calculate the median length for the 11 Aleppo pine needles.

Example: Aleppo Pine Needle Lengths

Ask Gemini: The pine data have two variables: `Pine` and `Length`. Get a subset for `Pine = Aleppo`.

```
# Create a subset for Aleppo pines
aleppo_subset <- subset(pine_data, Pine == "Aleppo")
```

Ask Gemini: For `aleppo_subset`, find the median of variable `Length`.

```
# Calculate the median
median_length_aleppo <- median(aleppo_subset$Length,
                              na.rm = TRUE)

# Print the result
median_length_aleppo
[1] 9.3
```

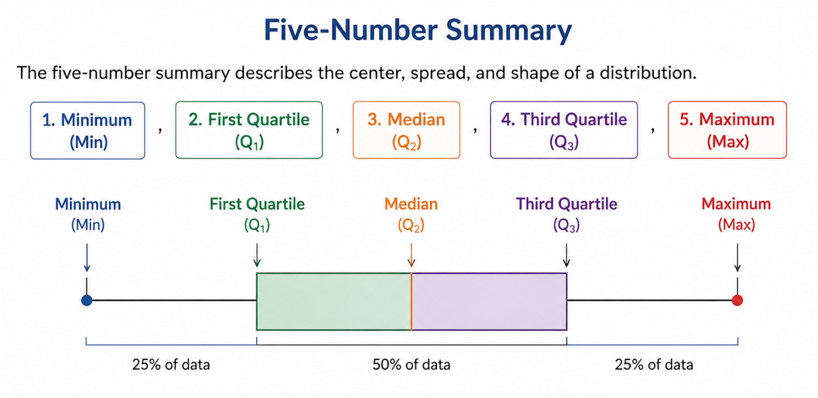
Comparing the Mean and Median

- For a symmetric distribution, the mean and median are close.
- For an exactly symmetric distribution, they are equal.
- In a skewed distribution, the mean is usually pulled farther into the long tail than the median.
- The mean is commonly used for symmetric data; the median is often preferred for skewed data.

Mean Versus Median

- The mean uses every observed value; the median depends mainly on order and position.
- The mean is affected by outliers, whereas the median is resistant.
- Use the mean to describe the center of approximately symmetric data.
- Use the median to describe the center of skewed data or data containing outliers.

Five-Number Summary



- Q_1 is the median of the observations below the overall median.
- Q_3 is the median of the observations above the overall median.

Interquartile Range and Detecting Outliers

The interquartile range measures the distance between the first and third quartiles:

$$\mathrm{IQR} = Q_3 - Q_1 .$$

The $1.5 \times \mathrm{IQR}$ **rule:** An observation is a suspected outlier if it is

$$\text{below } Q_1 - 1.5\mathrm{IQR} \quad \text{or above } Q_3 + 1.5\mathrm{IQR}.$$

Example: Five-Number Summary and IQR

Ask Gemini: For the `aleppo_subset`, find the five-number summary and compute the IQR.

```
# summary() provides Min, 1st Qu., Median,
# Mean, 3rd Qu., and Max
aleppo_summary <- summary(aleppo_subset$Length)

# Compute the IQR
aleppo_iqr <- IQR(aleppo_subset$Length, na.rm = TRUE)

# Print the results
print("Five Number Summary (+ Mean):")
print(aleppo_summary)
```

Example: Five-Number Summary and IQR

```
print(aleppo_summary)

   Min. 1st Qu. Median Mean 3rd Qu. Max.
  7.200 8.600 9.300 9.782 11.100 12.800

print("Interquartile Range (IQR):")
[1] "Interquartile Range (IQR):"

print(aleppo_iqr)
[1] 2.5
```

Thus, the five-number summary is

$$(7.2,\ 8.6,\ 9.3,\ 11.1,\ 12.8),$$

and

$$\mathrm{IQR} = Q_3 - Q_1 = 11.1 - 8.6 = \boxed{2.5}.$$

Measuring Spread: Variance and Standard Deviation

Variance measures spread by examining how far observations are from their mean:

$$s^2 = \frac{(x_1 - \bar{x})^2 + \cdots + (x_n - \bar{x})^2}{n-1} = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2.$$

The standard deviation is the square root of the variance:

$$s = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2}.$$

Properties of the Standard Deviation

- The denominator is $n - 1$, reflecting $n - 1$ degrees of freedom.
- Standard deviation measures spread around the mean.
- $s \geq 0$, and $s = 0$ only when there is no spread.
- Standard deviation has the same measurement units as the original observations.
- Like the mean, standard deviation is not resistant to outliers.

Example: Variance and Standard Deviation

Ask Gemini: Compute the variance and standard deviation of `Length` in `aleppo_subset`.

```
# 1. Compute the Variance
aleppo_variance <- var(aleppo_subset$Length,
                       na.rm = TRUE)

# 2. Compute the Standard Deviation
aleppo_sd <- sd(aleppo_subset$Length,
                na.rm = TRUE)

# Print the results
cat("Variance:", aleppo_variance, "\n")
Variance: 3.329636

cat("Standard Deviation:", aleppo_sd, "\n")
Standard Deviation: 1.824729
```

Interquartile Range as a Resistant Measure

- Variance and standard deviation are not resistant to outliers or strong skewness.
- In such cases, the IQR provides a resistant measure of spread.

$$\mathrm{IQR} = Q_3 - Q_1 .$$

Choosing Measures of Center and Spread

- For symmetric or approximately symmetric data, use the **mean** and **standard deviation.**
- For skewed data or data containing outliers, use the **median** and **IQR**.

Data Visualization

- Categorical variables: pie charts and bar charts
- Numerical variables: histograms and boxplots
- Relationships between two numerical variables: scatterplots and Pearson correlation

Health of the Nation

The National Center for Health Statistics (NCHS) is the principal health-statistics agency in the United States. Each year, it collects extensive information from a large number of households.

Questions of interest include:

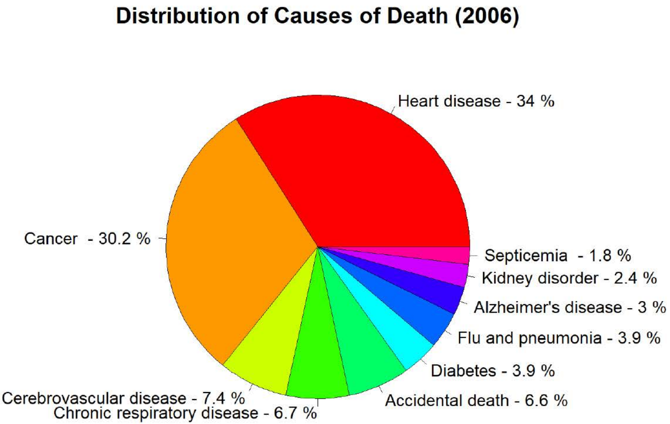
- What are the top 10 leading causes of death in the United States?
- What are the leading causes of death overall?

Top 10 Leading Causes of Death, 2006

Cause of death	Count	Percent
Heart disease	631,636	34%
Cancer	559,888	30%
Cerebrovascular disease	137,119	7%
Chronic respiratory disease	124,583	7%
Accidental death	121,599	7%
Diabetes	72,449	4%
Flu and pneumonia	72,432	4%
Alzheimer's disease	56,326	3%
Kidney disorder	45,344	2%
Septicemia	34,234	2%

Pie Charts

A pie chart displays the relative share of each category as a portion of a whole.



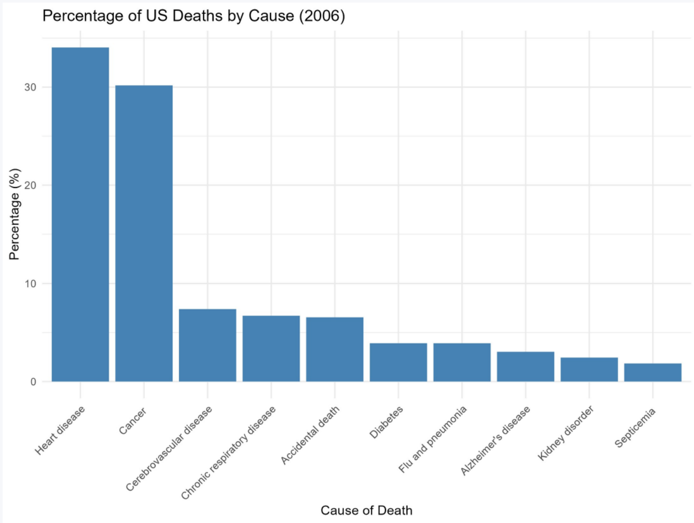
Example: Creating a Pie Chart

Ask Gemini: For the `death_data` data, write R code to make a pie chart of `Cause_of_death` according to its counts recorded in `Count`. Save the figure to `Pie.png`, make the label size bigger, and make the width 1800.

```
png(filename = "Pie.png", width = 1800,
   height = 1200, res = 150)
# Calculate percentages and labels
percent_labels <- round(100 * death_data$Count /
                  sum(death_data$Count), 1)
pie_labels <- paste(death_data$Cause_of_death, "-",
                  percent_labels, "%")
# Generate the pie chart
pie(death_data$Count,
   labels = pie_labels,
   main = "Distribution of Causes of Death (2006)",
   col = rainbow(nrow(death_data)),
   cex = 1.5,
   cex.main = 2.0)
# Save the file
dev.off()
```

Bar Charts

A bar chart displays category frequencies or percentages with separated bars.



Example: Creating a Bar Chart

Ask Gemini: Write R code with `ggplot2` to make a bar chart using the percentage of count, and make the bars vertical. Save it to `barchart.png`.

```
# Load the ggplot2 library
library(ggplot2)
# Calculate the percentage for each cause
death_data$Percentage <- (death_data$Count /
                         sum(death_data$Count)) * 100
# Create the vertical bar chart
p <- ggplot(death_data,
   aes(x = reorder(Cause_of_death, -Percentage),
       y = Percentage)) +
  geom_bar(stat = "identity", fill = "steelblue") +
  labs(title = "Percentage of US Deaths by Cause (2006)",
      x = "Cause of Death", y = "Percentage (%)") + theme_
          minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
# Save the chart
ggsave("barchart.png", plot = p,
     width = 8, height = 6, dpi = 300)
```

Histogram

What is the length of the great white shark?

The great white shark, *Carcharodon carcharias*, is a large ocean predator at the top of the food chain. Here are the lengths in feet of 44 great whites:

18.7	12.3	18.6	16.4	15.7	18.3	14.6	15.8	14.9	17.6	12.1
16.4	16.7	17.8	16.2	17.8	13.8	13.8	12.2	15.2	14.7	12.4
13.2	15.8	14.3	16.6	9.4	18.2	13.2	13.6	15.3	16.1	13.5
19.1	16.2	22.8	16.8	13.6	13.2	15.7	19.7	18.7	13.2	16.8

Make a histogram for this data set and comment on it.

Constructing a Histogram: Steps 1–3

1. Determine the minimum and maximum values:

$$\text{min} = 9.4, \qquad \text{max} = 22.8.$$

2. Choose the number of classes (bins), for example $k = 7$.
3. Determine a class width:

$$i \geq \frac{\text{max} - \text{min}}{k} = \frac{22.8 - 9.4}{7} = 1.91.$$

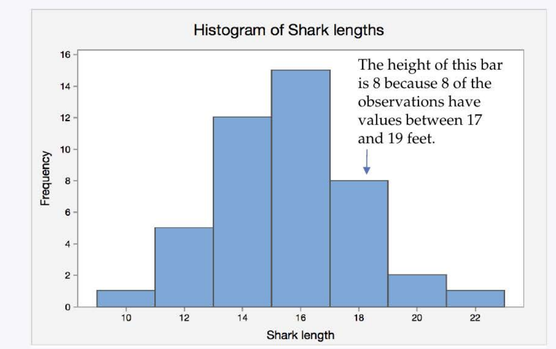
   A convenient class width is 2 feet.

Constructing a Histogram

Step 4: Count the individuals in each class (bin).

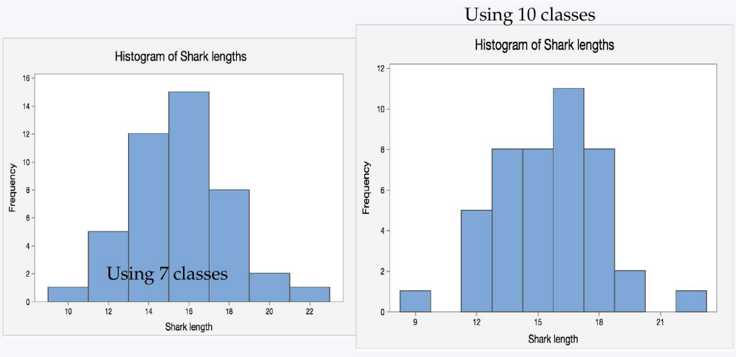
Class	Count
9 to 11	1
11 to 13	5
13 to 15	12
15 to 17	15
17 to 19	8
19 to 21	2
21 to 23	1
Total	**44**

Step 5: Draw the histogram using 7 classes.



Choosing the Number of Histogram Bins

- Different no. of bins can produce different impressions.
- Too few bins may hide important structure.
- Too many bins may exaggerate random variation.
- Statistical software usually provides a default or “optimal” binning rule, but the analyst should inspect alternatives.



Describing a Histogram

Look for the overall pattern and any striking deviations from that pattern.

Describe the distribution using:

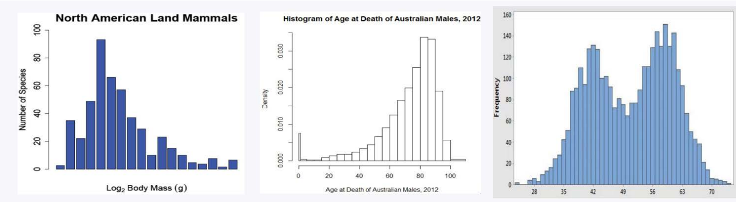
- **Shape**
- **Center**
- **Spread**
- **Outliers**

Histogram Shape

The shark-length distribution is approximately unimodal and fairly symmetric.

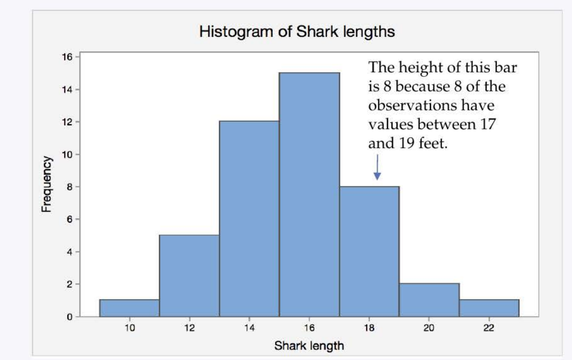
Other common shapes include:

- Unimodal and skewed to the right
- Unimodal and skewed to the left
- Bimodal
- Approximately symmetric

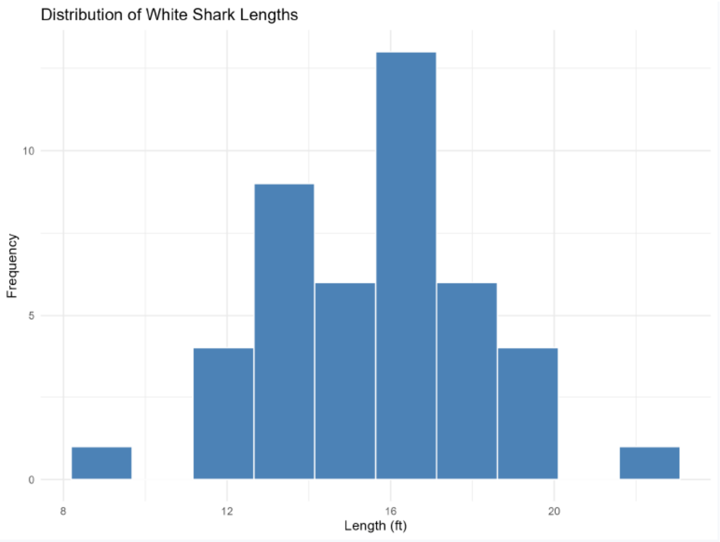


Histogram Center, Spread, and Outliers

- **Center:** The midpoint of the shark-length distribution is about 15–17 feet.
- **Spread:** Observed lengths range from 9.4 to 22.8 feet.
- **Outliers:** Isolated observations near the ends of the distribution should be investigated as possible outliers.



Plot from R + Gemini



Stemplot

To construct a stemplot:

1. Separate each observation into a **stem** and a final digit called the **leaf**.
2. Write the stems vertically in increasing order.
3. Write each leaf to the right of its stem, with leaves ordered from smallest to largest.

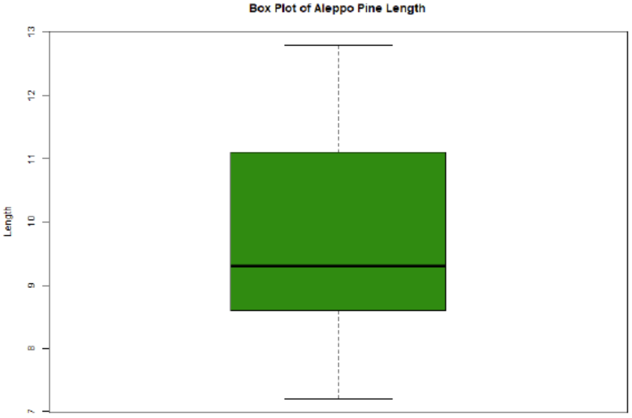
```
stem(shark_data$Length,scale=1)

# The decimal point is at the |
#
# 8 | 4
# 10 |
# 12 | 1234222256688
# 14 | 3679237788
# 16 | 122446788688
# 18 | 2367717
# 20 |
# 22 | 8
```

Boxplot

A boxplot is a graphical display of the five-number summary.

- The central box spans the first and third quartiles, Q_1 and Q_3.
- A line inside the box marks the median.
- Whiskers extend toward the smallest and largest non-outlying observations.
- Potential outliers are commonly displayed as individual points.

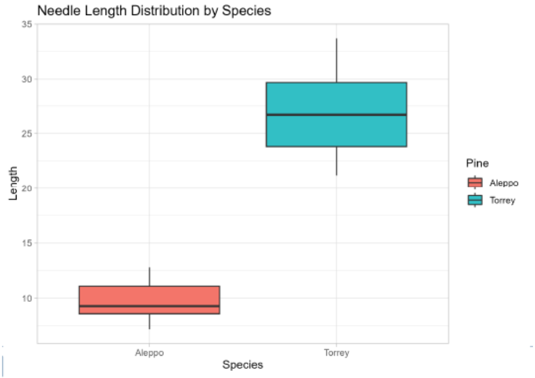
Box Plot for Aleppo Pine Needles



Comparing Groups with Side-by-Side Boxplots

For Aleppo and Torrey pine-needle lengths, compare:

- Center, using the median
- Spread, using the IQR and whisker lengths
- Skewness or asymmetry
- Potential outliers



Response and Explanatory Variable

So far, we have worked with only one variable and summarized it graphically and numerically. We now briefly cover the relationship between two quantitative variables, describe it graphically, and measure the association numerically.

- A **response variable** measures an outcome of a study. An **explanatory variable** may explain or influence changes in a response variable.

$$\textbf{Explanatory variable} \xrightarrow{\text{May affect}} \textbf{Response variable}$$

- **Examples:**
  - Number of beers and blood alcohol content
  - Age and an animal's weight

Displaying Relationships with a Scatterplot

A scatterplot displays the relationship between two quantitative variables measured on the same individuals.

- The explanatory variable is plotted on the horizontal axis.
- The response variable is plotted on the vertical axis.
- Each point represents one observational unit.

Example: Manatees and Powerboats

Manatees are large, herbivorous aquatic mammals found primarily in the rivers and estuaries of Florida.

A study examined the relationship between:

- The number of powerboats registered in a year
- The number of manatee deaths from powerboat collisions in that year

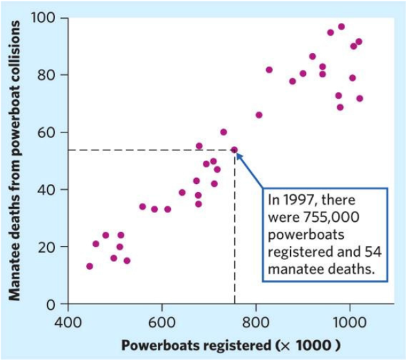
The data cover years from the late 1970s through the 2000s.

Powerboats and Manatee Deaths

Powerboat registrations (in thousands) and manatee deaths from powerboat collisions in Florida											
Year	Powerboats	Deaths	Year	Powerboats	Deaths	Year	Powerboats	Deaths	Year	Powerboats	Deaths
1977	447	13	1986	614	33	1995	713	42	2004	983	69
1978	460	21	1987	645	39	1996	732	60	2005	1010	79
1979	481	24	1988	675	43	1997	755	54	2006	1024	92
1980	498	16	1989	711	50	1998	809	66	2007	1027	73
1981	513	24	1990	719	47	1999	830	82	2008	1010	90
1982	512	20	1991	681	55	2000	880	78	2009	982	97
1983	526	15	1992	679	38	2001	944	81	2010	942	83
1984	559	34	1993	678	35	2002	962	95	2011	922	87
1985	585	33	1994	696	49	2003	978	73	2012	902	81

Powerboat registrations (in thousands) and manatee deaths from powerboat collisions in Florida

Interpreting the Manatee Scatterplot



The scatterplot shows annual manatee deaths against powerboat registrations, measured in thousands.
For example, in 1997 there were approximately 755,000 registered powerboats and 54 recorded manatee deaths, corresponding to the point $(755, 54)$.

Interpreting a Scatterplot

Examine the following characteristics:

- **Form:** Is the overall pattern linear or nonlinear?
- **Direction:** Is the association positive or negative?
- **Strength:** How closely do the points follow the overall pattern?
- **Outliers:** Are there observations that fall well outside the pattern?

Measuring Linear Association

Correlation measures the direction and strength of the linear relationship between two quantitative variables.
For paired observations (x_i, y_i), $i = 1, \ldots, n$,

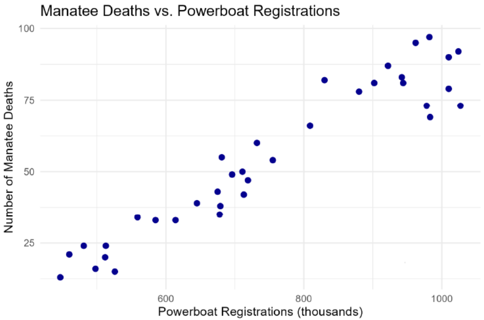
$$r = \frac{1}{n-1} \sum_{i=1}^{n} \left(\frac{x_i - \bar{x}}{s_x} \right) \left(\frac{y_i - \bar{y}}{s_y} \right).$$

The terms in parentheses are standardized values of x and y.

Facts About Correlation

- Correlation makes no distinction between explanatory and response variables.
- Because r uses standardized values, it does not change when measurement units are changed.
- Positive r indicates positive association; negative r indicates negative association.
- Correlation always satisfies $-1 \leq r \leq 1$.
- Values near zero indicate a weak *linear* relationship.

Example: Correlation




```
cor(manatee_data[,c("Powerboats","Deaths")])

# Powerboats Deaths
# Powerboats 1.0000000 0.9525368
# Deaths 0.9525368 1.0000000
```

The sample correlation is approximately $r = 0.953$.

Statistical Principles

- **Data vary.** Understanding variability is central to statistics; center alone does not adequately describe a distribution.
- **Always consider the distribution.** Numerical summaries should be interpreted together with graphical displays of the data.
- **The choice of summary depends on the data.** Use the mean and standard deviation for approximately symmetric distributions, and the median and IQR for skewed distributions or data with outliers.
- **Graphical choices matter.** Different graphs reveal different features of data, and choices such as histogram bin width can affect what we see.
- **Association is not causation.** A strong association between two variables does not, by itself, establish a causal relationship.